

ASSIGNMENT #3
Wireless and Mobile Networks Versions
Wireless and Mobile Network



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Analyze and compare Wireless and Mobile Networks Versions on the basis of data rate, applications(uses), modulation schemes, power consumptions. Present in a tabular form.

WIRELESS AND MOBILE NETWORKS COMPARSION

MOBILE NETWORK GENERATION(CELLULAR)

Table 1: TABLE OF MOBILE NETWORK GENERATION (CELLULAR)

GENERATION	DATA RATE	APPLICATION /USERS	MODULATION SCHEMES	POWER CONSUMPTION	KEY FEATURES
1G	2.4 kbps	Voice calls only, Analog voice transmission	FM (Frequency Modulation), FDMA (Frequency Division Multiple Access)	High(2-3W)	First mobile phone system, No digital capabilities, Poor voice quality
2G (GSM)	9.6-14.4 Kbps	Digital voice calls, SMS text messaging ,Basic data services	GMSK (Gaussian Minimum Shift Keying), TDMA (Time Division Multiple Access)	Medium (1-2W)	Digital encryption, Better voice quality, International roaming
2.5G (GPRS)	56-114 Kbps	Email, Web browsing, MMS messaging, Basic mobile internet	GMSK, Packet switching	Medium (1-2W)	Always-on connectivity, Pay-per-data usage, Bridge to 3G
2.75G (EDGE)	384 Kbps	Improved web browsing, File downloads, Video messaging ,Mobile TV	8-PSK (8-Phase Shift Keying), Enhanced GPRS	Medium (1-2W)	3x faster than GPRS, Better spectrum efficiency
3G (UMTS)	384 Kbps - 2 Mbps	Video calling, Mobile internet, Streaming media, Mobile TV, GPS navigation	QPSK (Quadrature Phase Shift Keying), W-CDMA (Wideband CDMA), CDMA2000	Medium-Low (0.5-1W)	Always-on broadband, Global roaming, Multimedia services

3.5G (HSPA)	5.76-42 Mbps	HD video streaming, Mobile broadband, Online gaming, VoIP services	16-QAM, 64-QAM, MIMO technology, Enhanced W-CDMA	Medium-Low (0.5-1W)	Faster data speeds, Improved latency, Better user experience
4G (LTE)	100 Mbps - 1 Gbps	HD/4K video streaming, Cloud services, Mobile gaming, IoT applications, Video conferencing	OFDMA (Orthogonal Frequency Division Multiple Access), SC-FDMA, 64-QAM, 256-QAM, MIMO	Low (0.2-0.5W)	All-IP network, Low latency, High spectral efficiency
4.5G (LTE-A)	1-3 Gbps	Ultra HD streaming, Advanced IoT, Real-time gaming, Augmented Reality	Carrier Aggregation, Enhanced MIMO, 256-QAM, 1024-QAM	Low (0.2-0.5W)	Multi-carrier support, Coordinated multipoint, Enhanced performance
5G	1-20 Gbps	Ultra HD/8K streaming, Autonomous vehicles, Industrial IoT, Virtual/Augmented Reality, Smart cities, Remote surgery	Massive MIMO, Beamforming, mmWave, OFDMA, LDPC coding	Very Low (0.1-0.3W)	Ultra-low latency (<1ms), Network slicing, Edge computing

WIRELESS LOCAL AREA NETWORK(Wi-Fi)

Table 2: TABLE OF WIRELESS LOCAL AREA NETWORK(Wi-Fi)

STANDARD	DAT A RATE	APPLICATION /USERS	MODULATION SCHEMES	POWER CONSUMPTION	KEY FEATURES
802.11 (Legacy)	2 Mbps	Basic networking, File sharing, Internet access	DSSS (Direct Sequence Spread Spectrum), FHSS (Frequency Hopping Spread Spectrum)	Medium (5-10W)	First wireless LAN standard, 2.4 GHz band
802.11b	11 Mbps	Internet browsing, Email, File transfers	DSSS, CCK (Complementary Code Keying)	Medium (3-8W)	Widely adopted, Good range,

					2.4 GHz band
802.11a	54 Mbps	Multimedia streaming, VoIP, Video conferencing	OFDM (Orthogonal Frequency Division Multiplexing)	Medium (3-8W)	5 GHz band, Less congested, Higher data rates
802.11g	54 Mbps	Streaming media, Online gaming, File sharing	OFDM, Backward compatible with 802.11b	Medium (3-8W)	2.4 GHz band, Combines best of a/b, Widespread adoption
802.11n (Wi-Fi 4)	150-600 Mbps	HD video streaming, Multi-device connectivity, Cloud backup	MIMO, OFDM, 64-QAM	Medium-Low (2-6W)	Dual-band (2.4/5 GHz), Multiple antennas, Channel bonding
802.11ac (Wi-Fi 5)	433 Mbps - 6.77 Gbps	4K video streaming, VR applications, High-density deployments	MU-MIMO, 256-QAM, Beamforming	Low (1-4W)	5 GHz only, Wider channels, Better efficiency
802.11ax (Wi-Fi 6)	600 Mbps - 9.6 Gbps	8K streaming, IoT devices, Smart homes, Enterprise applications	OFDMA, 1024-QAM, MU-MIMO, BSS Coloring	Very Low (0.5-3W)	Both 2.4/5 GHz, Target Wake Time, Improved efficiency
802.11be (Wi-Fi 7)	5.8-46 Gbps	Ultra-high bandwidth apps, Real-time applications, Metaverse/AR/VR	4096-QAM, Multi-Link Operation, Enhanced MU-MIMO	Very Low (0.5-3W)	6 GHz band support, Ultra-low latency, Maximum throughput

BLUETOOTH STANDARDS

Table 3:TABLE OF BLUETOOTH STANDARDS

Version	Data Rate	Applications /Users	Modulation Schemes	Power Consumption	Key Features
Bluetooth 1.x	721 Kbps	Basic file transfer, Audio headsets, Mouse/keyboard	FHSS, GFSK (Gaussian Frequency Shift Keying)	High (100mW)	10m range, Point-to-point, Basic security

Bluetooth 2.x (EDR)	2.1 Mbps	Enhanced audio, File transfers, Print services	$\pi/4$ -DQPSK, 8-DPSK, FHSS	Medium (10-100mW)	Enhanced Data Rate, Better security, Improved pairing
Bluetooth 3.x (HS)	24 Mbps	High-speed transfers, Video streaming, Large file sharing	Wi-Fi radio for data, Traditional BT for control	Variable	Wi-Fi integration, High-speed channel, Power optimization
Bluetooth 4.x (LE)	1 Mbps	Fitness trackers, IOT sensors, Wearables, Beacons	GFSK, Low Energy protocols	Very Low (0.01-0.5mW)	Ultra-low power, Coin cell battery life, Mesh networking
Bluetooth 5.x	2 Mbps	Smart home devices, Industrial IOT, Medical devices, Audio streaming	LE Coded PHY, Improved LE, Enhanced GFSK	Ultra-Low (0.01-1mW)	4x range, 2x speed, 8x broadcast capacity

CONCLUSION:

In conclusion, the power consumption values represent typical transmission power levels. The data rates represent theoretical maximum speeds that can be attained under optimal conditions. The modulation schemes describe the major digital modulation techniques that are employed. Actual performance can be influenced by environmental conditions, device specifications, and congestion in the network. Power consumption is also affected by the type of device, usage pattern, and implementation efficiency.